

SHANGRLA Assertion And Assorter Audit Contracts

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Abstract

SHANGRLA reframes many risk-limiting audits as tests of assertion means. An outcome claim is reduced to one or more assertions; each assertion supplies an assorter, a bounded function whose average must exceed one half under the reported outcome. This paper documents the RCOUNT V.16 surface that has landed: not a full SHANGRLA or ALPHA numerical replay, but a record-shape and design contract for assertion and assorter transcripts. The verifier accepts a named `shangrla-assorter-v1` audit run, checks assertion ids, registered assorter ids, bounds, sample-step links, p-values, and declared assorter values, and exposes failures through the common audit-algorithm transcript check. The contract makes the assertion layer reproducible before later work attaches a sequential mean test.

1 Introduction

SHANGRLA is useful because it changes the unit of an election audit claim. A contest outcome is no longer treated as one opaque stopping rule. It is reduced to assertions, and each assertion is evaluated by an assorter: a bounded value assigned to a ballot, CVR, comparison record, batch, or other sampled unit. When the assorter mean is greater than $1/2$, the assertion supports the reported outcome.

That abstraction is deliberately broader than any one audit method. BRAVO, Kaplan-Markov comparison audits, ALPHA betting martingales, and some ranked choice audit paths can all be described as tests over assertion-level evidence. The RCOUNT V.16 slice therefore focuses on the assertion and assorter contract itself. A package should say which assertion is being tested, which assorter is being used, which sample unit received which assorter value, and what declared risk evidence or decision is attached to that sequence.

The implemented claim is modest. RCOUNT does not yet recompute a SHANGRLA mean test from this method id, and it does not claim that a reported outcome has been risk-limited by V.16 alone. It verifies that the declared assertion and assorter record is reproducible as a public record shape: ids are linked, values are bounded, and the half-average layer is ready for a later ALPHA or other mean replay.

2 Assorter Contract

The landed record surface is the package-level `AuditAlgorithmRun`. For SHANGRLA the method id is the real RCOUNT constant `SHANGRLA_ASSORTER_METHOD_ID`, whose value is `shangr1a-assorter-v1`. The run contains one or more assertions and an ordered set of sample steps. Each step names the assertion it belongs to and the assorter value assigned to that sampled unit.

Field	Meaning
<code>run_id</code>	Stable id for the audit-algorithm transcript.
<code>contest_id</code>	Contest whose outcome assertion is being represented.
<code>method_id</code>	Versioned algorithm id; for this slice, <code>shangr1a-assorter-v1</code> .
<code>risk_limit_ppm</code>	Optional risk limit in parts per million, checked when present.
<code>assertion_id</code>	Stable assertion id, such as <code>assertion:cand-a-over-cand-b</code> .
<code>kind</code>	Assertion kind, including <code>plurality-winner-loser</code> or <code>assorter-mean</code> .
<code>assorter_id</code>	Registered assorter formula or function id, such as <code>plurality-winner-loser-v1</code> .
<code>assorter_upper_bound</code>	Positive rational bound for all values in the assertion.
<code>sample_unit_id</code>	Ballot, CVR, batch, or other sampled unit being assigned a value.
<code>assorter_value</code>	Nonnegative rational value used by the future mean test.
<code>statistic, p_value_ppm, decision</code>	Declared audit evidence and outcome fields; shape-checked here, not recomputed as SHANGRLA.

The half-average framing is the design boundary. The assertion is written so that the reported outcome is supported when the assorter mean exceeds one half of its normalized scale. The current verifier checks the declared bounded values that such a test would consume; it does not yet run ALPHA or any other sequential test under the SHANGRLA method id.

3 Implemented Surface

The method id is defined in `rcount-core` as `SHANGRLA_ASSORTER_METHOD_ID = "shangr1a-assorter-v1"`. The generic method registry accepts that id in `is_supported_audit_algorithm_method`. The design verifier is `verify_audit_algorithm_runs`; it is also the baseline package verifier surface that emits the pass id `audit_algorithm_transcript`.

The SHANGRLA slice is not a statistical replay branch in `rcount-audit`. The replay dispatcher has explicit branches for BRAVO, ALPHA, Minerva, Kaplan-Markov, batch comparison, and several documented boundary methods, but no SHANGRLA branch. A `shangr1a-assorter-v1` run therefore falls through to the generic boundary transcript, method does not yet have statistical replay. The real implemented guarantee is the core record-shape and design contract.

Check	Failure caught
<code>is_supported_audit_algorithm_method</code>	Unsupported method ids before any assertion record is trusted.
<code>risk_limit_ppm</code> in <code>verify_audit_algorithm_runs</code>	Zero or million-and-above risk limits when a risk limit is declared.
Duplicate run and assertion checks	Reused <code>run_id</code> or repeated <code>assertion_id</code> values.
Assertion-link check	Sample step naming an assertion that is not declared in the run.
Assorter-bound check	Empty <code>assorter_id</code> or nonpositive <code>assorter_upper_bound</code> .
Sample-step uniqueness	Repeated <code>assertion_id</code> , <code>step_index</code> pairs.
Assorter-value check	Negative values, nonpositive denominators, or values above the assertion bound.
Optional evidence checks	Invalid bet denominator, negative statistic, or <code>p_value_ppm</code> greater than 1,000,000.

Errors in this family are mapped by `rcount-audit` to the public check name `audit_algorithm_transcript_accepts_shangrla_algorithm_steps`. That keeps SHANGRLA boundary-design failures on the same transcript surface as the other audit algorithm records without claiming that the SHANGRLA p-value has been numerically replayed.

4 Fixtures

There is not yet a committed golden package named for V.16 in `crates/rcount-core/src/fixtures.rs`. That file contains several audit algorithm fixtures with assorter fields, including `synthetic_kaplan_markov_package`, `synthetic_minerva_round_one_package`, `synthetic_bayesian_tabulation_boundary_package`, `synthetic_ranked_choice_boundary_package`, and `synthetic_soba_observable_ballot_boundary_package`. Those are real assorter-bearing records, but they are not SHANGRLA golden fixtures.

The exact SHANGRLA fixture now present is the inline test helper `synthetic_shangrla_algorithm_run` in `crates/rcount-core/src/tests.rs`. It builds `audit-run:shangrla-toy` for `syn-2024-mayor` with method id `shangrla-assorter-v1`. The assertion is `assertion:cand-a-over-cand-b`, kind `PluralityWinnerLoser`, with assorter id `plurality-winner-loser-v1`, upper bound `1/1`, winner `cand-a`, and loser `cand-b`. Its two sample steps assign `cvr:P-001:001` value `1/1` and `cvr:P-001:002` value `1/2`.

The positive test is `audit_algorithm_transcript_accepts_shangrla_assorter_steps`. It attaches the helper run to `synthetic_summary_basic_package` and expects an `audit_algorithm_transcript_pass` for `audit-run:shangrla-toy`.

The negative tests are also shape tests. The test `audit_algorithm_transcript_rejects_missing_assertion` edits a sample step to name `assertion:missing` and expects `MissingAuditAssertion`. The test `audit_algorithm_transcript_rejects_out_of_bound_assorter_value` changes the first assorter value to `3/2` against an upper bound of `1/1` and expects `InvalidAuditAssorterValue`. The adjacent macro-design tests reuse the same helper after changing the method id to `Kaplan-Markov`; they are not SHANGRLA numeric replay fixtures.

5 Boundaries

The plan boundary is explicit: SHANGRLA defines the assertion layer. It does not replace source hashing, sampler replay, ballot manifests, or CVR linkage. A valid V.16 record can say which assertion and assorter values were declared; other RCOUNT contracts must still prove where the sampled records came from and whether the source bytes, manifests, CVRs, and public summaries are reproducible.

The implementation boundary is just as important. In the current code, `shangrila-assorter-v1` is a record-shape and boundary-design contract, not a full numeric replay. The core verifier checks ids, links, bounds, optional p-value ranges, and assorter values. The audit replay crate does not recompute a SHANGRLA assertion p-value for that method id. V.15 ALPHA can replay a bounded mean martingale for an ALPHA method id, but the plan still lists wiring ALPHA to consume explicit SHANGRLA assorter values as future work.

SHANGRLA also reduces contests to assertions; it does not by itself decide the correct assertion set for every election rule. A two-candidate plurality outcome can be represented as a winner-loser assertion. Multi-winner, approval, scoring, comparison, and ranked-choice audits require the reported outcome to be reduced to the right assertion family before the half-average test is meaningful. That reduction is the bridge V.16 prepares, not something the current verifier infers from arbitrary contest semantics.

6 Conclusion

RCOUNT now has the first SHANGRLA assertion and assorter record contract. The method id `shangrila-assorter-v1` is registered, audit algorithm runs can carry assertion records and bounded rational assorter values, and the verifier checks that sample steps link to declared assertions without exceeding their bounds. The current positive and negative tests show the intended public failure modes: a valid toy assertion passes the transcript-shape check, a missing assertion link fails, and an out-of-bound assorter value fails.

The result should be read narrowly. V.16 does not yet replay SHANGRLA risk math or certify a contest outcome. It establishes the reproducible assertion layer that later ALPHA, comparison, and ranked-choice work can consume. That is the right order: make the declared assertion and assorter record inspectable first, then attach the statistical replay that turns those values into a stopping claim.

References

- Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026.
- Philip B. Stark. Shangrila: Sets of half-average nulls generate risk-limiting audits. Technical report, arXiv, 2020.